Framing Financial Feedback: Narrative vs. Factual AI Explanation in an Educational Simulation

Margaret Yan[1], Emily Regan[1], Wendy Wang[1]
[1]Human-Computer Interaction Lab, Department of Computer Science, Wellesley College, Wellesley, MA, 02481

BACKGROUND & SIGNIFICANCE

- Students with limited financial literacy face barriers to relatable education, and traditional programs rely on abstract, one-size-fits-all curricula that do not adapt to lived experience [1,2]
- How something is explained matters as much as what is explained, a principle that treats audiences as active learners but predates AI-generated feedback [3]
- Serious games produce knowledge gains through consequence-driven mechanics [1,2,4,5]
- Narrative framing improves comprehension and perceived trust over feature-based explanations [6]; prior knowledge shapes how critically users evaluate them [7]
- Explanation style inside a serious game has received less attention than the mechanics around it

RESEARCH OBJECTIVES

- **RQ1:** How does narrative versus factual framing in AI-generated feedback affect how college students comprehend and internalize financial literacy concepts?
- **RQ2:** What does explanation form reveal about learners' relationship to the content?

METHODOLOGY

- Study Design
  - Mixed-methods, quasi-experimental, between-subjects
  - Participants randomly assigned to one condition (*N = 20, n = 10 per condition*)
- Quantitative data collection & analysis:
  - Pre- and post-study quizzes to test for concept comprehension before and after the study, compared between conditions in SPSS
  - Debrief responses tallied by condition across three closed questions
- Qualitative data collection & analysis:
  - Mid-study comprehension free, written responses inductively coded into themes by all three authors
  - Themes compared across conditions and across three levels

ACKNOWLEDGEMENTS

We would like to thank our professor, Dr. Orit Shaer, for her guidance and feedback throughout the development of this work. We also thank our teaching fellow, Becky Chen, for her continued support and assistance.

GAME DESIGN



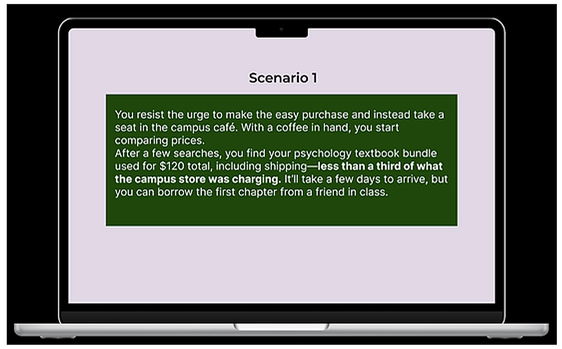


Figure 1 (left). Figma prototype of one scenario and its answer choices in narrative condition on desktop view

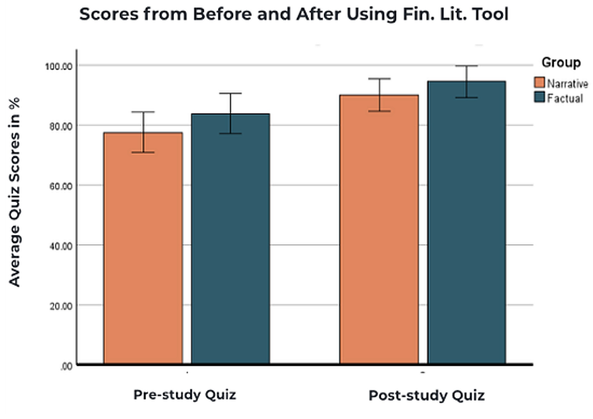
Figure 2 (right). Figma prototype of one scenario and its explanation in narrative condition on desktop view

- Desktop-based game with three levels covering budgeting, debt management, and saving
- Two configured GPT models generated the feedback, framed in either a narrative or factual condition
- Scenarios and explanations finalized in advance in an interactive Figma prototype to eliminate live-prompt variance

Narrative Condition
- Sequential, story-like reflections
- Outcome values woven into story
- Plain, everyday language

Factual Condition
- Concise, data-driven feedback
- Outcome values stated directly
- Technical financial terminology

QUANTITATIVE RESULTS



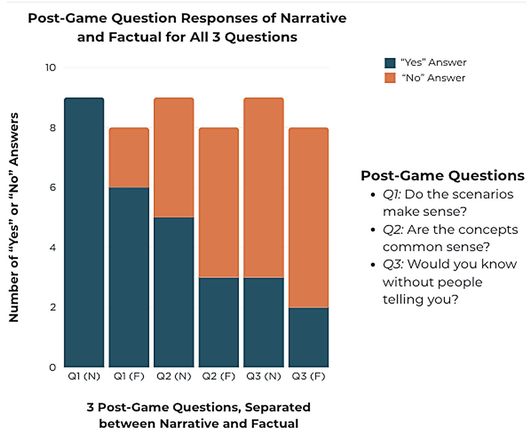


Figure 3 (left). Bar chart showing no statistically significant improvements in financial knowledge between baseline and retention surveys for both narrative and factual conditions

Figure 4 (right). Stacked bar chart showing the number of "yes" and "no" responses during the post-study verbal debrief for both narrative and factual conditions

- Random assignment left factual group about 6 points higher at baseline, so conditions were compared on gains rather than post-study scores
- Scores improved from pre-game to post-game by about 12 points in narrative condition and by 10 points in factual condition
- The 2-point difference in knowledge gain between conditions was not statistically significant
- 5 of 9 narrative participants called the concepts common sense, against 3 of 8 in the factual condition

QUALITATIVE RESULTS

- Delayed gratification was the takeaway across all three levels.
- Debt management (Level 2) had the most specific technical learning, with students naming new knowledge (interest rates, credit card mechanics).
- Themes recurred across both conditions, suggests that experiential structure supports reflection regardless of explanation mode.
- Participants often noted building emergency funds while balancing discretionary spending.

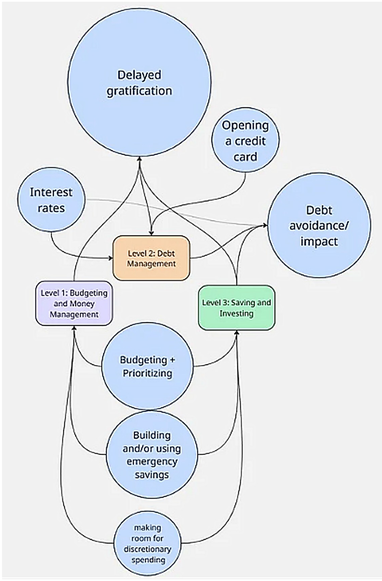


Figure 5. Themes from mid-study written responses on concepts, linked to the levels where they appeared

DISCUSSION

- Gains and themes were alike across conditions, suggesting experiential structure rather than explanation style drove learning
- Explanation style may shape what learners treat as already known rather than how much they learn, a pattern not statistically supported
- Isolating surface language from content makes explanation form a testable communication design variable

LIMITATIONS AND FUTURE WORK

- Twenty students from one selective liberal arts college; high baseline financial exposure may have created a ceiling effect
- Some participants found certain scenarios lacked a clear right answer, which may have made takeaways harder for some participants to articulate
- Individual differences between small groups may explain the qualitative pattern; Need for Cognition and the Narrative Transportation Scale to be tested as moderators
- Retention tested one week later to assess longer-term effects of explanation style
- Deploy as a web app with a live financial dashboard, conversational AI advisor, and what-if toggle for exploring counterfactual outcomes

REFERENCES

[1] Hu et al. (2023). Financial Decision Buddy: A Decision-Support Tool to Bridge the Gaps in Financial Education. CHI EA '23.
[2] Santos et al. (2025). DreamScape: Fostering Financial Cognition through a Serious Game. C&C '25.
[3] Mehlenbacher (2008). Communication Design and Theories of Learning. SIGDOC '08.
[4] Weichselbraun et al. (2025). Cash Action: An Educational Multiplayer Game for Teaching Financial Concepts. FDG '25.
[5] Chen (2025). Research on the Application of Large Language Model to Financial Digital Literacy Education. Int'l Conf. on Information Education and AI.
[6] Martens et al. (2025). Tell Me a Story! Narrative-Driven XAI with Large Language Models. Decision Support Systems.
[7] Celar & Byrne (2023). How People Reason with Counterfactual and Causal Explanations for AI Decisions in Familiar and Unfamiliar Domains. Memory & Cognition.
[8] Aryal & Keane (2023). Even if Explanations: Prior Work, Desiderata & Benchmarks for Semi-Factual XAI. arXiv.
[9] Cacioppo & Petty (192). The Need for Cognition. Journal of Personality and Social Psychology.
[10] Green & Brock (2000). The Role of Transportation in the Persuasiveness of Public Narratives. Journal of Personality and Social Psychology.